

1) Since the system is causal, initial conditions are 0, we take Laplace

$$s^3 Y(s) + (1+\alpha)s^2 Y(s) + \alpha(\alpha+1)s Y(s) + \alpha^2 Y(s) = X(s)$$

$$Y(s) [s^3 + (1+\alpha)s^2 + \alpha(\alpha+1)s + \alpha^2] = X(s)$$

$$H(s) = \frac{1}{s^3 + (1+\alpha)s^2 + \alpha(\alpha+1)s + \alpha^2} = \frac{1}{(s+\alpha)(s^2+s+\alpha)}$$

s+α parenthesis

a)

$$G(s) = sH(s) + H(s) = (s+1)H(s) = \frac{s+1}{(s+\alpha)(s^2+s+\alpha)}$$

If $\alpha \neq 1$, $s+\alpha=0 \rightarrow s=-\alpha$ (1 pole)
 $s^2+s+\alpha=0$ (2 poles) } 3 poles

If $\alpha = 1$, $G(s) = \frac{1}{s^2+s+1} \rightarrow 2$ poles

b)

For stability $s = -\alpha$
 $-\alpha < 0 \rightarrow \alpha > 0$

$s^2+s+\alpha=0$ $As^2+Bs+C=0$
 From Routh-Hurwitz $A=1 > 0 \checkmark$
 $B=1 > 0 \checkmark$

in order to system to be stable both must meet the conditions.

Common state is $\alpha > 0 \checkmark$

$C = \alpha \rightarrow$ must be > 0

2) a. $\mathcal{L}\{e^{-at} u(t)\} = \frac{1}{s+a}$ ROC: $\text{Re}\{s\} > -a$

$$X(s) = \frac{1}{s+1} \text{ ROC: } \text{Re}\{s\} > -1$$

$$H(s) = \frac{1}{s+2} \text{ ROC: } \text{Re}\{s\} > -2$$

b) $Y(s) = X(s) \cdot H(s) = \frac{1}{(s+1)(s+2)}$ ROC: $\text{Re}\{s\} > -1$

örnek Laplace

$$\frac{A}{s+1} + \frac{B}{s+2} \Rightarrow A = \frac{1}{s+2} \Big|_{s=-1} = \frac{1}{-1+2} = 1$$

$$B = \frac{1}{s+1} \Big|_{s=-2} = \frac{1}{-2+1} = -1$$

$$Y(s) = \frac{1}{s+1} + \frac{1}{s+2}$$

$$y(t) = \mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} = \underline{e^{-t}u(t) - e^{-2t}u(t)}$$

3) $h[n] = a^n u[n]$

$$H(z) = \frac{1}{1-az^{-1}}, \text{ ROC: } |z| > |a|$$

$$x[n] = 1$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} = \sum_{n=0}^{N-1} 1 \cdot z^{-n} = \sum_{n=0}^{N-1} (z^{-1})^n \rightarrow \text{geometrik seri}$$

$$X(z) = \frac{1 - (z^{-1})^N}{1 - z^{-1}} = \frac{1 - z^{-N}}{1 - z^{-1}}, \text{ ROC: } |z| > 0$$

$$Y(z) = X(z) \cdot H(z) = \left(\frac{1 - z^{-N}}{1 - z^{-1}}\right) \left(\frac{1}{1 - az^{-1}}\right)$$

$$Y(z) = \frac{1}{(1 - z^{-1})(1 - az^{-1})} - \frac{z^{-N}}{(1 - z^{-1})(1 - az^{-1})} \rightarrow V(z) = \frac{1}{(1 - z^{-1})(1 - az^{-1})}$$

$$Y(z) = V(z) - z^{-N}V(z) \rightarrow \text{kısmi kesirler ayırma ile şöyle bulunur}$$

$$y[n] = \underline{v[n] - v[n-N]}$$

$$y[n] = \frac{1}{1-a} (1 - a^{n+1}) u[n]$$

$$- \frac{1}{1-a} (1 - a^{n-N+1}) u[n-N]$$

$$\frac{1}{1-a} \left[\underbrace{\frac{1}{1-z^{-1}}}_{u[n]} - \underbrace{\frac{a}{1-az^{-1}}}_{a \cdot a^n u[n] = a^{n+1} u[n]} \right]$$

$$v[n] = \frac{1}{1-a} (1 - a^{n+1}) u[n]$$

$$4) a. Y(z) = z^{-1} Y(z) + z^{-2} Y(z) + z^{-1} X(z)$$

$$Y(z)(1 - z^{-1} - z^{-2}) = z^{-1} X(z) \quad H(z) = \frac{z^{-1}}{1 - z^{-1} - z^{-2}} = \frac{z}{z^2 - z - 1}$$

$$\text{Zeros: } z = 0$$

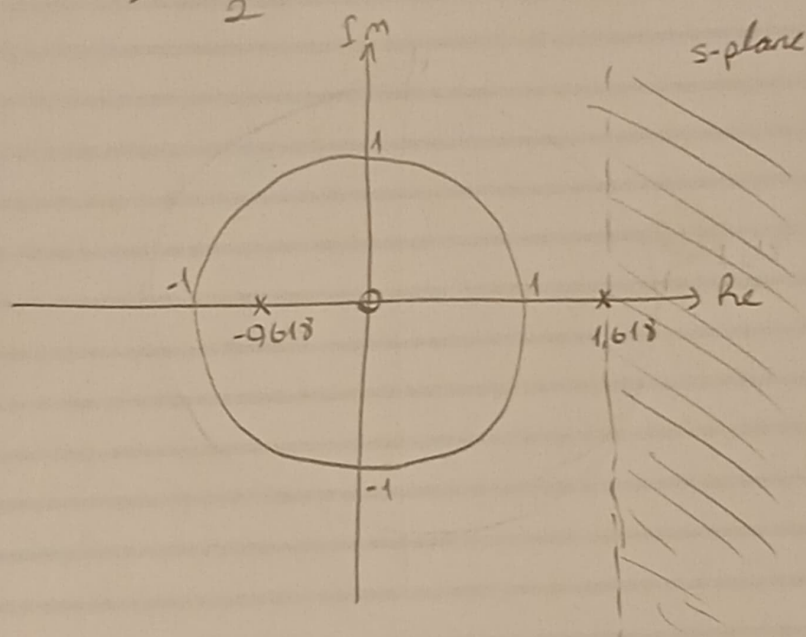
$$\text{Poles: } z^2 - z - 1 = 0$$

$$z_{1,2} = \frac{1 \pm \sqrt{5}}{2}$$

$$z_1 = 1,618$$

$$z_2 = -0,618$$

$$\text{ROC: } |z| > \frac{1+\sqrt{5}}{2} (=1,618)$$



b.

$$H(z) = \frac{z^{-1}}{(1 - z_1 z^{-1})(1 - z_2 z^{-1})} = \frac{A}{1 - z_1 z^{-1}} + \frac{B}{1 - z_2 z^{-1}}$$

$$z_1 = \frac{1+\sqrt{5}}{2} \quad z_2 = \frac{1-\sqrt{5}}{2} \quad \rightarrow A = \frac{1}{\sqrt{5}} \quad B = -\frac{1}{\sqrt{5}}$$

$$H(z) = \frac{1}{\sqrt{5}} \left[\frac{1}{1 - z_1 z^{-1}} - \frac{1}{1 - z_2 z^{-1}} \right]$$

$$h[n] = \frac{1}{\sqrt{5}} [(z_1)^n - (z_2)^n] u[n] = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right] u[n]$$

c.

$$|z_2| = |-0,618| = 0,618 < 1 \quad (\text{Kutup içinde } \checkmark)$$

$$|z_1| = |1,618| = 1,618 > 1 \quad (\text{Kutup dışında})$$

Because of this the system is unstable